

July

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2008	July 18	July 3
2009	July 7	July 22
2010	July 26	July 11
2011	July 15	July 1, 30
2012	July 3	July 19

ECLIPSES

2009: July 21–22 A total eclipse of the Sun is visible from India, China, and the Pacific Ocean. A partial solar eclipse is seen from eastern and southeastern Asia, Indonesia, and the Pacific Ocean.

2010: July 11 A total eclipse of the Sun is visible from the southern Pacific Ocean.

THE PLANETS

2008: July 1

Mercury is at greatest morning elongation, magnitude 0.6.

2008: July 9

Jupiter is at opposition, magnitude -2.7 . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2008: July 10

Mars and Saturn appear about a Moon-width apart in the western evening sky.

2010: July 30–31

Mars and Saturn appear about four Moon-widths apart in the western evening sky.

2011: July 20

Mercury is at greatest evening elongation, magnitude 0.5.

2012: July 1

Mercury is at greatest evening elongation, magnitude 0.5.

2012: July 1–2

Venus and Jupiter appear about ten Moon-widths apart in the eastern dawn sky.

Opposition and elongation are explained on [page 14](#)

Hercules and Ophiuchus remain center stage, and Aquila, the eagle, has flown into view. Southern observers are treated to the rich star fields of the Milky Way. These include Sagittarius, within which, almost directly overhead, lies the center of our galaxy. The outstretched wings of Cygnus, the swan, are now high in the sky for northern observers. Vega, in the small constellation, Lyra, shines brightly overhead.

NORTHERN LATITUDES

Looking north, Ursa Minor reaches up from Polaris. Coiling around it is the long, winding figure of Draco, the dragon. Ursa Major is to the left of Polaris, in the northwest; to the right, in the northeast, are the mythical king Cepheus and his queen, Cassiopeia. Arcturus (Boötes), in the west, and Vega (Lyra), almost overhead, are the brightest stars in the July sky.

Next in brightness is Altair in the southeast, marking the neck of Aquila, the eagle, and flanked by two dimmer stars. The Milky Way flows through Aquila and on to Cygnus, high in the east. A line of stars forms the swan's body and an intersecting line its wings; this cross shape led to its other name, the northern cross. Cygnus's brightest star, the blue-white supergiant Deneb, marks its tail.



Vega

The blue-white star Vega (below center) is the brightest star in the northern summer sky. Close by is the star Epsilon (ϵ) Lyrae. Sharp eyesight or binoculars will show it as a double star (upper left). A telescope reveals that each of its two stars is also a double.

One of the smallest constellations, Corona Borealis is high in the southwest; it is flanked by Hercules and Boötes. Its arc of seven stars depicts the crown of the mythical Princess Ariadne, hence its alternative name, the Northern Crown.

Ophiuchus is well placed due south, with Hercules above.

Observers south of about 45°N will see star-rich

Sagittarius and Scorpius on their horizons. This is the best chance for northern observers to see the two most southerly constellations of the zodiac in the evening sky.

